Claudespeak: Characterizing the Stylistic Fingerprint of a Frontier Language Model

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Abstract

Large language models are widely believed to have recognizable writing styles, but most popular claims (e.g. that models overuse the word *delve*) are derived from *ChatGPT* and applied indiscriminately to all models. We ask what, specifically, characterizes the prose of one frontier model—Anthropic's Claude Opus 4.8—relative to human writing and to eight other models, holding topic constant via four parallel corpora that span informational QA, essays, real in-the-wild prompts, and ten professionally-authored task types. Combining interpretable stylometry, log-odds n-gram selection, a length-controlled regression, an interpretable classifier, and suffix-automaton (n-gram novelty) mining, we find a stable fingerprint that is *structural, rhythmic, and interactional* rather than a matter of stereotyped vocabulary. The fingerprint replicates with consistent sign across all four corpora, and Claude is identifiable from prose alone in every one (prose-only classifier AUC 0.79–0.95 across four independent contrasts), driven by markedly higher sentence-length variability, denser content-word usage, a strong em-dash habit, and—most diagnostically—a verbatim offer-to-continue closing move ("would you like me to go deeper into…") that, in our corpus, appears dozens of times in Claude and *zero* times across humans and all other models. We further show that the stereotyped "delve" vocabulary is a GPT (and other-model) trait that Claude *avoids*: across seven models in an essay genre, Claude is the only model that never writes *delve* and uses *crucial* least. Finally, we audit corpus coverage and, finding the standard benchmarks narrow, broaden it with two diverse sources, releasing the fully provenanced corpus and pipeline. Sections on the effect of reasoning effort, Claude versions, and multi-turn dynamics are forthcoming.

1 Introduction

The claim that text "sounds like AI" is now commonplace, and a folk vocabulary of tells—*delve*, *tapestry*, *crucial*, *intricate*—circulates widely. Two problems undercut these claims as characterizations of any *particular* model. First, the evidence base is overwhelmingly about ChatGPT (Kobak et al., 2025), yet the conclusions are applied to "LLMs" generically. Second, most analyses do not control topic: a classifier that distinguishes model from human text may be learning the prompt distribution, not the writing style.

We study a single frontier model, Anthropic's Claude Opus 4.8, and ask a deliberately narrow question: *holding the prompt fixed, what makes Claude's prose identifiable relative to humans and to other models?* We build two topic-matched parallel corpora in which the same prompts are answered by Claude, by humans, and by up to six other models, and we attack the question with five complementary methods. Our contributions:

1. A parallel-corpus protocol and a fully provenanced dataset—four tracks, eight contrast models and two human anchors, every generation stored with model, decoding configuration, token usage, and timestamp—that supports reuse without regeneration.

2. Evidence that Claude's fingerprint is *structural and rhythmic* (sentence-length burstiness, content-word density, em-dashes), survives both markdown stripping and length control, and *replicates with consistent sign across all four corpora* despite different prompt distributions and contrast sources.

3. Identification of a verbatim *interactional* signature—an offer-to-continue closer—via log-odds and suffix-automaton mining, present in Claude and absent from all other sources in our corpus.

4. A reconciliation of the "delve" folklore: the stereotyped vocabulary is a GPT/other-model trait; Claude is the cleanest frontier model on it.

5. A *coverage taxonomy* (intent $\times$ interaction structure $\times$ stance) and a reproducible audit showing that standard corpora occupy a narrow slice (single-turn, cooperative, informational). We then *act on* the audit, broadening coverage with two diverse sources—real in-the-wild prompts and ten professionally-authored task types—and show the fingerprint persists, while naming the cells that remain under-observed (multi-turn corrective, contested-stance) as targets for future work. We release the audit tooling.

2 Related Work

Detecting machine-generated text. A large literature aims to decide *whether* a text is machine-generated. Supervised detectors fine-tune classifiers such as RoBERTa on model outputs (Solaiman et al., 2019), while zero-shot methods exploit statistical signatures of the generating distribution: probability curvature (Mitchell et al., 2023), its efficient conditional-sampling successor (Bao et al., 2024), structured search over weaker-model features (Verma et al., 2024), or the cross-perplexity of a model pair (Hans et al., 2024). "In-the-wild" benchmarks (Li et al., 2024b; Wang et al., 2024) and Dugan et al. (2024) show such detectors are brittle across domains and attacks; paraphrasing in particular evades them (Krishna et al., 2023), with adversarial training (Hu et al., 2023) as a partial defense and arguments that reliable detection may be fundamentally hard (Sadasivan et al., 2023). Watermarking (Kirchenbauer et al., 2023) sidesteps detection by injecting a signal at generation time. We do not detect machine text; we instead *characterize* the style of one frontier model, assuming provenance is known.

Attributing text to a model. Closer to our aim is work on *which* model wrote a text. A recent forensic survey organizes this space into detection, attribution, and *characterization* (Kumarage et al., 2024); the last is our category, and the problem is also framed as authorship attribution in the LLM era (Huang et al., 2024). Concrete attribution methods exploit a range of signals—multi-class benchmarks over many generators (Uchendu et al., 2021), contrastive perplexity for origin tracing (Li et al., 2023a), information density (Venkatraman et al., 2024), and learned style representations (Rivera Soto et al., 2024)—and, closest to us, interpretable stylometry (Kumarage and Liu, 2023). One result matters for our framing in particular: a model's stylistic signature persists even in text deliberately optimized to evade detectors (Rivera Soto et al., 2025), so style is robust enough to be worth characterizing in its own right. Rather than optimize a label, we use an interpretable, length-controlled regression and log-odds n-gram selection (Monroe et al., 2008) to surface *what* distinguishes Claude, and we audit which stylistic claims hold up under corpus coverage.

Characterizing model writing style. Most directly related is recent work that compares the *style* of LLM and human text rather than detecting it. Reinhart et al. (2025) build parallel human/LLM corpora and apply Biber's grammatical/rhetorical features, finding systematic, scale-growing differences; Sun et al. (2025) show models are mutually distinguishable from word-level idiosyncrasies that survive paraphrase; and McGovern et al. (2024) recover model "fingerprints" from simple lexical/POS frequencies. Classic stylometry has likewise been applied to AI creative writing (O'Sullivan, 2025), and a parallel line operationalizes low-quality "AI slop" as measurable dimensions (Shaib et al., 2025). We share the parallel-corpus, interpretable-feature stance but go deeper on a single model: we characterize Claude specifically, isolate an *interactional* closing move that feature inventories do not capture, add suffix-automaton novelty evidence, and show the fingerprint replicates across four corpora.

Excess vocabulary and "AI-ese." Kobak et al. (2025) quantify LLM-marker words (e.g. "delve") as excess vocabulary in the biomedical literature, and corpus-scale estimators track the rising LLM footprint in papers, peer reviews, and society at large (Liang et al., 2024b,a), with parallel evidence from arXiv abstracts (Geng and Trotta, 2024) and even spoken language (Yakura et al., 2024). Popular accounts trace the habit to RLHF annotation pipelines (Willison, 2024), a hypothesis given peer-reviewed support by Juzek and Ward (2025). Such word-lists, however, are largely a ChatGPT/GPT signature and are unstable over time—"delve" fell once it was publicly flagged (Geng and Trotta, 2025). We therefore treat the "delve" folklore as a hypothesis to test rather than a given: our markers are discovered data-drivenly from parallel corpora, and we show this trait is *not* characteristic of Claude.

Stylometry and n-gram novelty. Our features build on classical stylometry, notably function-word frequencies and Burrows's Delta (Burrows, 2002). To probe memorization (Carlini et al., 2023) versus genuine novelty, we adopt suffix-automaton n-gram search (Merrill et al., 2024a,b), unbounded n-gram indexing (Liu et al., 2024), and related corpus-search tooling (Elazar et al., 2024), applying novelty mining to a single model's outputs comparatively. We read novelty as relative distinctiveness, not creativity: it correlates with but does not equal creative quality (Lu et al., 2024; Saakyan et al., 2026).

How post-training shapes style. The voice we characterize is a product of post-training. Instruction tuning and RLHF (Ouyang et al., 2022), and for Claude RL from AI feedback (Bai et al., 2022), produce the helpful "assistant" and its register, which can be read as a trained persona (Shanahan et al., 2023) with steerable, training-induced trait directions (Chen et al., 2025), and which Anthropic describes as deliberately shaped by character training (Anthropic, 2024). Crucially, several of our specific markers have documented post-training causes: reward correlates with response length (Singhal et al., 2023), preference models favor lists, bold, and emphasis (Zhang et al., 2024), and RLHF reduces output diversity (Kirk et al., 2024; Padmakumar and He, 2024). This profile—verbose, formatted, and homogenized—matches the markdown-scaffolded, content-dense voice we measure, though we test for the style itself rather than its mechanism.

Behavioral signatures and multi-turn dynamics. Beyond surface style, frontier models show measurable behavioral signatures: verbosity bias in LLM judges (Zheng et al., 2023), sycophancy and agreement under pressure (Perez et al., 2022; Sharma et al., 2023; Xie et al., 2024), exaggerated-safety over-refusal (Röttger et al., 2024), persona and identity drift across turns (Li et al., 2024a; Choi et al., 2024), and self-interaction attractors such as the "spiritual bliss" state reported for Claude (Anthropic, 2025). Most of these are interaction-gated, and they motivate our future-work agenda on multi-turn corrective behavior, where a stylistic fingerprint may interact with concession and drift dynamics.

Data. We draw on parallel human/LLM corpora including HC3 (Guo et al., 2023) and AlpacaEval (Li et al., 2023b), and broaden coverage with real in-the-wild user prompts from WildChat (Zhao et al., 2024) and the professionally hand-authored No Robots dataset (Rajani et al., 2023), repurposing all of them for comparative stylometric characterization rather than detection or preference ranking. Table 1 situates these against the broader landscape of public corpora usable for comparative LLM stylometry, noting for each which sources (human and which models) it contains, whether multiple sources answer the *same* prompt (the parallel property our design needs), whether it is multi-turn, and what annotations it carries. Two properties are scarce in combination: most corpora are either parallel-but-narrow (HC3, AlpacaEval) or diverse-but-not-parallel (WildChat, LMSYS-Chat-1M); Chatbot Arena and MT-Bench (Zheng et al., 2024, 2023) are naturally parallel and multi-turn and are the most promising material for future extensions.

3 Corpus

We construct four parallel tracks (Table 2). In each, a fixed set of prompts is answered by multiple sources, so that any feature separating Claude reflects *how* it writes rather than *what* it was asked. Claude (and GPT-4o on the HC3 track) are generated by us with adaptive thinking at `effort=high`; all other cells are reused from the source datasets. Every record stores full provenance—model and version, decoding configuration, token usage, timestamp, and code commit—plus the verbatim API response.

The first two tracks (HC3, AlpacaEval) supply a human anchor and a modern multi-model contrast but, as our coverage audit shows (§3.2), they occupy a narrow behavioral slice: short, single-turn, information-seeking *questions*. The last two tracks are added precisely to broaden that slice. **WildChat** contributes real, in-the-wild first-turn user prompts (heavy on creative writing, code, and roleplay) with the original GPT-4-0314 reply as a reused contrast. **No Robots** contributes professionally hand-written prompts *and* answers, balanced across ten task types (generation, brainstorming, rewriting, summarization, classification, extraction, coding, chat, and open/closed QA); it

Dataset	Size*	Sources (human / models)	Parallel†	Turns	Key annotations	Used
HC3	24k Q	human; ChatGPT (GPT-3.5)	H vs model	single	human/AI label	✓
AlpacaEval	805 instr.	6+ LLMs (no human)	cross-model	single	GPT-4 preference / win-rate	✓
WildChat	1M conv	real users; GPT-3.5/4 (incl. 0314)	no	multi	toxicity, language, redaction, geo	✓
No Robots	10k	human prompts & answers	human only	single	task category (×10)	✓
LMSYS-Chat-1M	1M conv	25 LLMs (no human)	no	multi	toxicity, language, model id	cand.
Chatbot Arena	33k conv	20+ LLMs, pairwise	2/prompt	multi	human preference vote, language	cand.
MT-Bench	80 Q ×6	6 LLMs	cross-model	multi (2)	expert pairwise judgments	cand.
TuringBench	168k	19 generators + human	partial	single	human/machine + author label	—
MAGE	448k	27 LLMs + human	no	single	detection + generator id	—
M4	122k	multi-gen, multilingual	no	single	detection + attribution	—
RAID	6M+	11 models × attacks	no	single	detection labels	—

Table 1: Landscape of public corpora for comparative LLM stylometry (cited in §2 as they appear: HC3, AlpacaEval, WildChat, No Robots, LMSYS-Chat-1M / Chatbot Arena / MT-Bench, TuringBench, MAGE, M4, RAID). The four tracks we use (top) are complementary; candidate ("cand.") corpora are promising for future extensions; the rest are detection/attribution benchmarks useful mainly for external validation. *Sizes approximate (prompts, conversations, or items as native to each). †*Parallel* = multiple sources answer the same prompt (the property that controls topic in our design); "no" = one model per conversation.

Track	Prompts	Sources (same prompt each)
HC3	200	human, ChatGPT (GPT-3.5 era), GPT-4o†, **Claude**†
AlpacaEval	200	gpt-4-turbo, gpt-4o, gemini-pro, deepseek-67b, Qwen2-72B, Llama-3-70B, **Claude**†
WildChat	1500	GPT-4-0314, **Claude**†‡
No Robots	1500	human (professional), **Claude**†

Table 2: The four parallel corpus tracks. † = self-generated; all other cells reused. HC3 and No Robots carry human anchors; AlpacaEval carries the modern multi-model contrast; WildChat carries real in-the-wild prompts. ‡ Claude on WildChat is currently a 628/1500 subset (generation interrupted); all WildChat comparisons use the matched 628-prompt subset, and the remaining cells will be filled and the tables regenerated.

Track	Prompts	Prompt wds (md)	Claude wds (md)	Total words
HC3	200	10	267	156k
AlpacaEval	200	18	230	354k
WildChat	1500‡	35	334	946k
No Robots	1500	49	162	397k

Table 3: Corpus volume and length (median word counts). Totals sum all sources in the track. ‡ Claude answered 628 of the 1500 WildChat prompts so far. Full per-source distributions: Appendix B.

is our second human anchor and the first one on non-QA tasks. Across all tracks there are eight distinct non-Claude models (GPT-3.5-era ChatGPT, GPT-4o, GPT-4-turbo, GPT-4-0314, Gemini-pro, DeepSeek-67B, Qwen2-72B, Llama-3-70B) and two human anchors (informal web text in HC3, professional annotators in No Robots).

3.1 Corpus statistics

Table 3 summarizes volume and length. The four tracks span a wide range of registers and lengths. Prompts grow from terse HC3/AlpacaEval questions (median 10–18 words) to substantial WildChat and No Robots tasks (median 35 and 49 words). Response lengths likewise spread: Claude's median answer ranges from 162 words (No Robots' often-constrained tasks) to 334 on WildChat, where its creative-writing responses push the mean to 611; the two human anchors are short and variable (HC3 median 77, No Robots median 48). We account for length via length controls (§5.3). Per-domain volume and full distributions are in Appendix B. A corpus-level register signal already foreshadows the results: Claude's responses are the most *interrogative* at the sentence level on every track (17%/10%/6%/9% of sentences on HC3/AlpacaEval/WildChat/No Robots, vs. ≤5% for every contrast source—the offer-closer, §5.5), and Claude is denser than its human anchors (Flesch ≈40–48).

3.2 Coverage: auditing and expanding the behavioral slice

Because style is context-dependent, we audit *what behavior the prompts elicit*, labeling every prompt by intent, speech act, register, tone, and topic, plus interaction flags (`src/analyze_coverage.py`; large tracks sampled at n=400). Labels are produced by Claude Opus 4.8 itself (the model under study), a circularity we flag in Limitations.

TODO (human validation pending): hand-annotate a ∼50-prompt sample, report inter-annotator agreement with the LLM labels, and add a small validation table. Quick human pass to be added.

The audit shows that HC3 and AlpacaEval alone occupy a narrow region: HC3 is 96% explanation/QA/advice and 75% *questions*; AlpacaEval is broader but still question/directive and information-oriented. Treating this as a coverage problem to fix rather than merely report, we add the WildChat and No Robots tracks, and the four tracks are *complementary* (Table 4): WildChat is dominated by creative writing (42%) and code (11%), No Robots by classification/ extraction (25%), summarization/rewriting (19%), and QA (20%). Intents that were absent or near-absent in the first two tracks—creative writing, code, roleplay, summarization, rewriting, classification, extraction, and personal/emotional—are now substantially represented. The expansion also shifts the other axes (Appendix B): the dominant speech act flips from HC3's 75% *questions* to 59–73% *directives* on the new tracks; a non-trivial *formal* register appears (10% of WildChat prompts) where it was nearly absent; and topics broaden from HC3's health/science/finance concentration to include entertainment, software, the arts, and personal life.

Two limits remain, and we are explicit about them. First, the corpus is still *single-turn*: only WildChat shows a non-trivial fraction of prompts that imply follow-up (7%), and essentially none are adversarial or corrective, so behaviors gated on a multi-turn *declarative* challenge—e.g. the concession move "you're right to push back on that"—remain out of scope and motivate the multi-turn extension (§5.9). Second, contested-stance prompts (those that hand Claude an assertion to agree with or contest) are still a small minority on every track. The fingerprint we report should therefore be read as Claude's *single-turn* voice—but now across a markedly broader range of single-turn tasks than informational QA alone.

Intent (primary, %)	HC3	Alpaca	WildC	NoRob
explanation	47	17	9	4
factual QA	27	15	11	20
advice/recommendation	25	16	5	9
instruction task	2	13	8	2
creative writing	0	12	**42**	10
code	0	6	11	10
summarize/rewrite/edit	0	4	3	**19**
classification/extraction	0	5	1	**25**
opinion/subjective	0	6	2	0
math/roleplay/other	1	11	9	2
flag: multi-turn impl.	4	2	7	1
flag: subjective	7	20	11	8
flag: sensitive	31	8	20	6

Table 4: Prompt intent distribution and interaction flags, % of audited prompts (HC3/AlpacaEval n=200; WildChat/No Robots sampled n=400; `coverage_labels.csv`). The four tracks are complementary: bold marks each new track's distinctive contribution. Multi-turn and contested-stance prompts remain scarce on all tracks.

4 Methods

We compute interpretable features per response (lexical: type-token ratio, hapax rate, function-word rate; punctuation: em-dash, colon, question, exclamation, emoji; syntax: mean sentence length and *burstiness*, the ratio of the standard deviation to the mean of words-per-sentence; markdown/structure; and rhetoric). Rates are per 100 or 1000 words so that text length does not mechanically dominate. We compute two dozen such features per response. We summarize source differences with Cohen's d (Claude vs. a comparison group); all reported effects are significant at $p < 0.001$ (Welch's t, n=200 Claude vs. $\geq$200 comparison). The classifier is ℓ_2-regularized logistic regression (C=1.0, balanced class weights). To separate *voice* from *formatting*, we additionally compute all prose features after stripping markdown markup. We mine characteristic n-grams with the Monroe et al. (2008) log-odds estimator; quantify length-independence with OLS (§5.3); measure separability with ℓ_2-regularized logistic regression under 5-fold cross-validation; and mine arbitrary-length spans and n-gram novelty with a suffix automaton (Merrill et al., 2024a).

5 Results

5.1 A large signal, dominated at first by formatting

The strongest raw separators between Claude and the other HC3 sources are markdown-structural:

Prose feature	$\lvert d\rvert$	Claude / others
Sentence burstiness	1.69	0.86 / 0.36–0.55
Function-word rate ↓	1.47	0.33 / 0.37–0.44
Em-dash rate	0.87	0.88 / ≤0.40
Colon rate	0.84	1.41 / 0.24–1.69
Question rate	0.82	0.61 / 0.01–0.28

Table 5: HC3 track, Claude vs. pooled others, markdown stripped. Mean |Cohen's d| and group means (Claude / range over the other sources). ↓ = Claude lower.

markdown headers (d=2.9), bold (d=2.3), and bullets (d=1.8), which the human and old-ChatGPT cells use essentially never. This is real but is a *formatting* habit; the decisive question is whether anything survives removing it.

5.2 The voice survives markdown stripping

With all markup stripped, Claude still separates strongly on prose (Table 5): it interleaves short and long sentences far more than any other source (burstiness), writes denser content-word prose (lower function-word rate), and retains a marked em-dash habit. These are the load-bearing stylistic findings.

5.3 The signature is not a length artifact

On the HC3 track, Claude writes longer answers (mean 253 words vs. 120 for humans, 228 for GPT-4o), so we test whether the prose effects merely track length. Within every word-count quartile the effects persist, and a length-controlled regression (HC3) (feature $\sim$ is_claude + z(n_words)) leaves the Claude coefficient large and highly significant: burstiness t=20.6, function-word rate t=−14.4, em-dash t=12.6, question rate t=4.6. Type-token ratio is length-sensitive and we treat lexical-diversity claims as length-conditioned. The effects also hold *within every domain* (Appendix C): across all nine domains of the HC3 and AlpacaEval tracks, burstiness has $d > 0$, function-word rate $d < 0$ (Claude denser), and em-dash and question rate $d > 0$, so the fingerprint is not a topic artifact; the next section extends this check to two further corpora.

5.4 The fingerprint replicates across all four tracks

The prose and length analyses above are on HC3. Do the same features distinguish Claude when the prompt distribution and the contrast source change? We compute Claude-vs-pooled Cohen's d separately within each track (for WildChat, on the matched 628-prompt subset, preserving the parallel-corpus control; `src/analyze_tracks.py`). The headline features carry the *same sign in all four tracks* (Table 6): Claude writes with higher sentence-length burstiness, denser content-word prose (lower function-word rate), more em-dashes, more markdown scaffolding, and more questions—against a human panel, against six modern models, against GPT-4-0314 on real prompts, and against professional human writers on diverse tasks. Magnitudes are largest on HC3, whose short, informal contrast cells make the gap widest, and shrink on the harder, more varied contrasts, but the direction never flips. (Two weaker features— colon rate and type-token ratio—are *not* sign-stable across tracks and we do not claim them as part of the fingerprint.) The interactional question habit is the one headline feature that fades on No Robots (d=0.03), whose classification, extraction, and rewriting tasks do not invite an offer to continue—a sensible boundary condition rather than a failure (§5.5).

Feature	HC3	Alp	WiC	NoR	sign
sentence burstiness	1.56	0.45	0.58	0.63	+4/4
function-word rate	−1.29	−0.57	−0.49	−0.24	−4/4
em-dash rate	0.86	0.12	0.48	0.34	+4/4
markdown headers	4.06	1.81	1.26	0.66	+4/4
markdown bold	2.72	0.88	1.12	1.09	+4/4
question rate	0.35	0.40	0.32	0.03	+4/4

Table 6: Claude-vs-pooled Cohen's d within each track (raw features). The sign is consistent across all four tracks for every headline feature; magnitude is largest where the contrast is cleanest (HC3). The question/offer habit is robust except on No Robots, whose task mix does not invite continuation.

5.5 The interactional signature: an offer-to-continue closer

Data-driven n-gram mining (Monroe et al., 2008) surfaces a clear interactional fingerprint rather than a content-word list. The most Claude-like trigrams (HC3) are *would you like*, *like me to*, *me to explain*, *in more detail*, *to go deeper*; the anti-signature—terms Claude *underuses*—is led by the formal frame *it is important to*. At the sentence level this closer registers as a striking jump in interrogatives: **17%/10% of Claude's response sentences are questions (HC3/AlpacaEval), versus ≤4% for every other source** (Appendix B).

Prompt (ELI5): *Why is skin itchy?*

Human: "As far as I know, scratching is an evolved reaction to help stop insects and other parasites from sucking our blood…"

Claude Opus 4.8: "…scratching sends a *new* signal to your brain that covers up the itchy feeling. But scratching too much can hurt your skin, so it's better to be gentle! **Would you like me to explain anything else?**" [closes with an emoji]

Figure 1: Same prompt, two sources (excerpts). Claude's answer is longer, more structured, and ends with the offer-to-continue closer; the human answer is terse and declarative.

per-1k	Claude	g4t	g4o	qwen	llama
delve	**0**	.018	.018	0	.017
delves	**0**	.037	.089	.019	.034
crucial	**.039**	.422	.340	.346	.185
intricate	**0**	0	0	.019	.050
pivotal	**0**	.073	.018	.038	0

Table 7: Frequency (per 1000 words) of stereotyped "excess" words on the AlpacaEval essay track. g4t = gpt-4-turbo, g4o = gpt-4o. Claude is at or near the minimum for every term. (gemini/deepseek omitted for space; same pattern.)

Suffix-automaton mining (§5.8) confirms the pattern at arbitrary length. Figure 1 shows the fingerprint in a single matched pair.

5.6 The "delve" vocabulary is a GPT trait Claude avoids

On the AlpacaEval track—an essay/instruction genre that *does* elicit the stereotyped words—Claude is the cleanest of seven models (Table 7). It is the only model that never writes *delve* or *delves*, and it uses *crucial* least of all seven (0.039 per 1k vs. DeepSeek 0.054, Gemini 0.090, Llama 0.185, GPT-4o 0.340, Qwen 0.346, GPT-4-turbo 0.422). Claude's own lexical lean is conversational ("actually" occurs ∼6× more often in Claude than GPT-4o on HC3); this conversational lean is a secondary signal beneath the structural one.

5.7 Separability

An interpretable classifier confirms a stable, learnable voice (Table 8). Stripping markdown reduces AUC only modestly, so the classifier is not merely reading layout; and its top coefficients are exactly the features of §5.2 (burstiness, function-word density, em-dash, question rate), an independent-method confirmation. Raw length is not a feature, and residualizing every feature on response

Track (contrast)	w/ fmt	prose
HC3 (human, GPT-4o, 3.5)	0.982	0.946
AlpacaEval (6 modern)	0.901	0.867
WildChat (GPT-4-0314)	0.973	0.854
No Robots (human, pro)	0.922	0.788

Table 8: Claude-vs-rest ROC-AUC (logistic regression, 5-fold CV) on all four tracks. Claude is highly separable under every contrast, including from prose alone; the lowest prose-only AUC (No Robots, 0.79) is the hardest case—against professional human writers on short, constrained tasks—and is still well above chance.

length barely changes separation (prose-only AUC 0.946→0.931 on HC3, 0.867→0.861 on AlpacaEval), so the result is not a length artifact.

5.8 Arbitrary-length spans and n-gram novelty

Suffix-automaton mining (Merrill et al., 2024a) gives the most concrete form of the signature. Within each track, spans up to *eight words* occur dozens of times in Claude and *zero* times across that track's non-Claude sources (HC3: humans + 2 models; AlpacaEval: 6 models, no human): on HC3, *would you like me to explain any* (×20) and *would you like me to go deeper into* (×14); on AlpacaEval, *would you like me to expand on any* (×7). Claude also has the highest n-gram novelty of any source (Figure 2; e.g. bigram novelty 0.519 vs. 0.27–0.36 for the six AlpacaEval models): its exact phrasing overlaps least with the pool. We read novelty as relative distinctiveness, not absolute creativity (see Limitations).

5.9 Secondary analyses (in progress)

Three planned analyses are not yet included; designs and placeholder tables appear in Appendix D.

- **Effect of reasoning effort/thinking.** Whether the fingerprint intensifies or attenuates across `effort`∈ {low, medium, high} and with thinking disabled. *[To be completed; Appendix D.1.]*
- **Across Claude versions.** Intra-Claude drift across model versions (e.g. Opus/Sonnet/Haiku and older releases). *[To be completed; Appendix D.2.]*
- **Multi-turn and self-interaction.** Turn-dependent traits and the Claude-to-Claude

Figure 2: **[Placeholder image—replace with real plot.]** n-gram novelty by source: the fraction of each source's n-grams (n=1 . . . 10) that are absent from the pooled other sources (AlpacaEval track). Claude's curve sits above all six other models at every n (bigram novelty 0.519 vs. 0.27–0.36), i.e. its phrasing overlaps least with the pool. Data: `reports/ngram_dawg.md`.

Replace the grey placeholder with the real novelty-curve figure (one line per source).

"spiritual bliss" attractor reported in (Anthropic, 2025). *[To be completed; Appendix D.3.]*

6 Discussion

Across four corpora, five methods, eight contrast models, and two human anchors, the same fingerprint recurs and survives every control: Claude-speak is markdown-scaffolded, rhythmically varied, content-dense, and engagement-and-offer-oriented prose—and specifically *not* the "delve" register. That the signature holds in sign from terse factual QA to in-the-wild creative writing to constrained extraction tasks suggests it is a property of the model rather than of any one prompt genre. A methodological lesson follows: borrowed AI-word lists largely measure *GPT*, and characterizing a specific model requires data-driven discovery against topic-matched references.

Limitations

We study a single Claude configuration (Opus 4.8, `effort=high`); the effort, version, and multi-turn axes are not yet measured (§5.9). Our coverage expansion (§3.2) broadens the corpus well beyond informational QA—into creative writing, code, summarization, rewriting, classification, and extraction—but it remains *single-turn*: context-gated behaviors that require a multi-turn declarative challenge (corrective concession, sustained refusal) are still out of scope and are the target of the planned multi-turn extension (English only). The WildChat Claude cells are currently a 628/1500 subset (generation interrupted); all WildChat comparisons use the matched subset, and the remaining cells and refreshed tables are pending. Reused contrast completions are their published vintages (e.g. GPT-4-0314, gemini-pro 1.0, Qwen2-72B, GPT-4-turbo/4o 2024), modern-ish but not the absolute latest. n-gram novelty is measured relative to our corpus pool, not a large external reference, and may be inflated for Claude as the lone out-of-cluster (and only freshly generated) source. The two human anchors differ in register—HC3 is informal web text, No Robots is professional annotation—which lets us check that the human–Claude gap is not merely an artifact of informal writing (the fingerprint holds against both), though neither is edited published prose. The coverage audit's labels are produced by an LLM (the model under study), introducing a circularity; a human-validation pass on a labeled sample is pending. Finally, raw length is excluded from the classifier features and the feature effects are length-controlled (§5.3), but some prose features still correlate with length; we report a length-residualized classifier as a check (§5.7).

Data and Reproducibility

We do not test mechanism, but the interactional, low-formality, offer-oriented profile is consistent with post-training (RLHF) that rewards helpful, engaging, and appropriately hedged responses. Code and the fully provenanced corpus (generations plus metadata, and the audit tooling) will be released at an anonymized URL. HC3, AlpacaEval, WildChat, and No Robots are public research artifacts used under their licenses; we redistribute only derived statistics and our own model generations.

Ethics Statement

This work characterizes model writing style using public datasets and model outputs; it involves no human subjects or private data. Stylistic fingerprints can aid provenance and detection but could also inform evasion; we report aggregate, interpretable features rather than a deployable detector.

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A Feature definitions

Full definitions of all interpretable features (rates, burstiness, markdown and rhetorical markers) are provided with the released code (`src/features.py`).

B Full corpus statistics

All statistics here are reproducible via `src/dataset_stats.py` (volume, lengths, sentence mood, register) and `src/analyze_coverage.py` (intent).

Response length by source (words, mean/median). HC3: human 120/77, ChatGPT 176/178, GPT-4o 228/239, Claude 254/267. AlpacaEval: deepseek-67b 186/184, gemini-pro 222/234, Claude 254/230, Qwen2-72B 260/259, gpt-4-turbo 272/285, gpt-4o 280/285, Llama-3-70B 297/308. WildChat: GPT-4-0314 375/357, Claude 611/334 (the high mean is driven by long creative-writing responses). No Robots: human 90/48, Claude 175/162.

Volume by domain / task (prompts; total response words, all sources). HC3: general 80/47k, finance 40/40k, cs_ai 40/35k, medicine 40/33k. AlpacaEval: selfinstruct 64/79k, oasst 49/96k, helpful_base 35/70k, koala 35/67k, vicuna 17/43k. No Robots (balanced, 150 prompts each): generation, brainstorm, coding, and rewrite are the highest-volume task types; classify, extract, and closed-QA the lowest (short answers). WildChat carries no source task labels (real user logs); its task mix is the LLM-labeled distribution in Table 4.

Sentence mood (% of response sentences). Interrogative sentences: Claude 17%/10%/6%/9% (HC3/AlpacaEval/WildChat/No Robots) vs. 1–5% for every contrast source—Claude is the most interrogative on every track. Imperative and exclamatory sentences are $\leq$2% and $\leq$7% respectively throughout.

Register. First-/second-person pronoun rates per 100 words and Flesch Reading Ease are in `stats_register.csv`. Claude is denser (lower Flesch) than both human anchors (HC3 human 48, No Robots human 59, vs. Claude ≈40–48) and than the modern-model panel on AlpacaEval; on WildChat its readability is comparable to GPT-4-0314's.

Prompt variety (audit). Where HC3 and AlpacaEval are narrow—almost all *questions/directives*, overwhelmingly *neutral* tone, and nearly no *formal* register—the two added tracks broaden every axis (Table 9): directives become the dominant speech act, a non-trivial formal register appears on WildChat (10%), and topics spread from HC3's health/science/finance concentration to entertainment, software, the arts, and personal life. Non-neutral tone and contested-stance assertions remain scarce on all four tracks.

Coverage expansion (delivered, and remaining). The WildChat and No Robots tracks (§3.2) populate intents that HC3/AlpacaEval lacked: creative writing, code, roleplay, summarization, rewriting, classification, and extraction. Two cells remain under-observed and motivate future work: (i) multi-turn corrective scenarios (a user challenges or corrects Claude) to elicit concession / push-back behavior, and (ii) a small, defensively-framed sensitive/refusal set. The audit tooling re-runs unchanged on any added track.

Axis	Value	HC3	Alp	WiC	NoR
speech act	question	75	53	23	38
	directive	23	47	74	59
	assertion	0	0	2	0
register	neutral	65	65	62	79
	casual	35	31	27	19
	formal	0	3	10	1
tone	neutral	82	88	86	90
	emotional	10	2	1	1
	playful	0	9	11	7

Table 9: Prompt variety (% of audited prompts) along *pragmatic* speech act (LLM-labeled), register, and tone. HC3/AlpacaEval n=200; WildChat/No Robots sampled n=400. The added tracks shift the dominant speech act to directives and introduce a formal register; non-neutral tone and assertions stay rare. Topic and intent breakdowns: `coverage_labels.csv`. (Syntactic mood, which differs, is in §3.2.)

Track	Domain	burst.	func↓	em-d.	ques.
HC3	cs/ai	2.50	−1.86	1.12	1.02
HC3	finance	2.11	−1.75	0.94	1.55
HC3	general	1.46	−0.97	0.95	0.17
HC3	medicine	2.65	−1.28	2.41	0.72
Alpaca	helpful	1.19	−0.81	0.66	0.85
Alpaca	koala	0.99	−0.43	2.02	0.25
Alpaca	oasst	1.46	−0.77	1.38	1.26
Alpaca	selfinst.	0.56	−0.40	0.26	0.29
Alpaca	vicuna	1.98	−1.17	3.22	2.21

Table 10: Per-domain Claude-vs-others Cohen's d, prose only. burst. = sentence burstiness; func↓ = function-word rate (negative = Claude denser); em-d. = em-dash; ques. = question rate. Uniform signs across domains.

C Per-domain robustness

Claude-vs-pooled-others Cohen's d (markdown-stripped prose) computed *within* each domain (Table 10). The signs are uniform across all nine domains—burstiness and em-dash/question positive, function-word rate negative (Claude denser)—so the fingerprint is not driven by any single topic.

D Placeholders for secondary analyses

D.1 Effect of reasoning effort

Planned design. Regenerate the 200 HC3 prompts (and 200 AlpacaEval prompts) under `effort`$\in$ {low, medium, high} and with thinking disabled, holding all else fixed; recompute the §5.2 features and the §5.7 classifier within each setting; report whether burstiness, density, em-dash, and the offer-closer scale with effort. *[Results table to be inserted.]*

D.2 Across Claude versions

Planned design. Generate the same prompts with additional Claude versions (e.g. Sonnet/Haiku, and an older 3.x), then measure intra-Claude drift in the fingerprint features and the offer-closer rate. *[Results table to be inserted.]*

D.3 Multi-turn and self-interaction

Planned design. (i) Two-turn parallel probes (e.g. MT-Bench-style) to measure turn-dependent traits comparatively; (ii) elicited Claude-to-Claude conversations to test the "spiritual bliss" attractor (Anthropic, 2025) quantitatively (e.g. trajectory of consciousness/gratitude vocabulary and vocabulary collapse over turns). *[Results table to be inserted.]*